



Safe & Clean

Classrooms

**Module 1: Water Resources &
Accessibility**

Version 2.0 (March 2026)



Our Learning

Goals

- 💧 Common **water sources** in your community
- 💧 What are family and community water sources, and how can you judge their **reliability**?
- 💧 Understand the **basic standards** for drinking water and other consumption

Let's get started!

Topic 1

**Water
Sources**

Water in the World



**What water sources and facilities
can you spot in this image?**

Do you see any challenges?

Water Source Classification

1 Rainwater

Water that has fallen as rain, as a result of water vapor evaporation and condensation



2 Surface water

Water that is collected on the surface

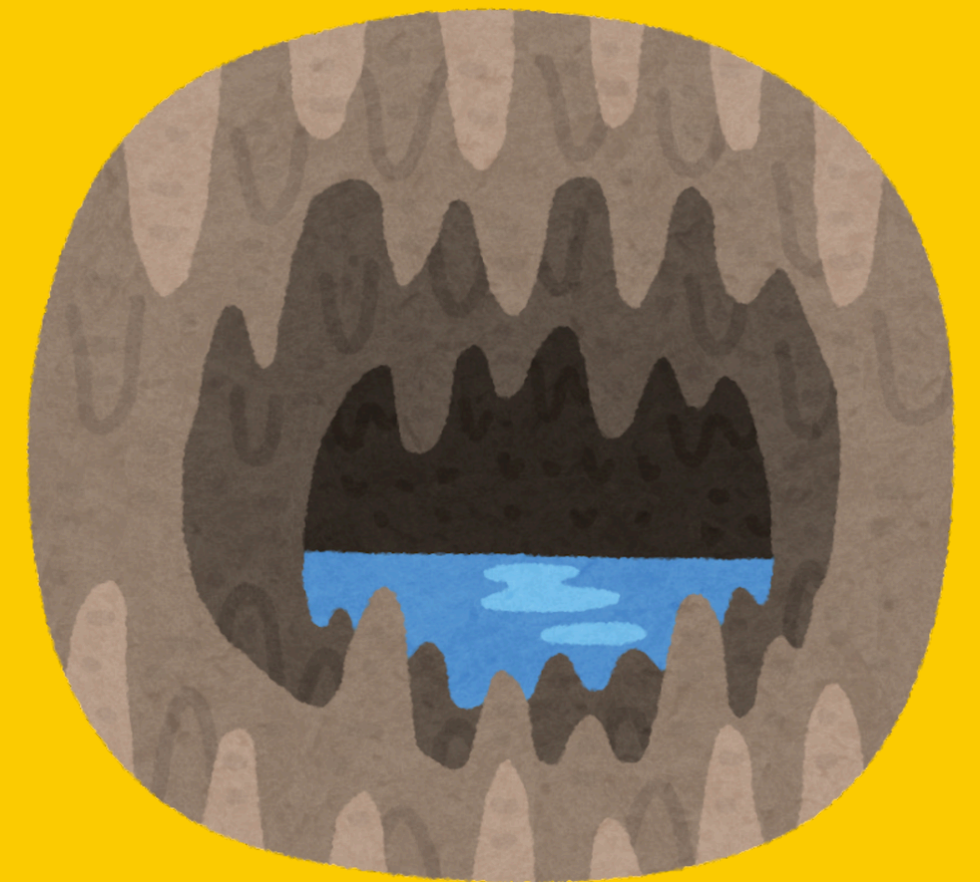


Sources can include: seas, oceans, rivers lakes, dams, etc.

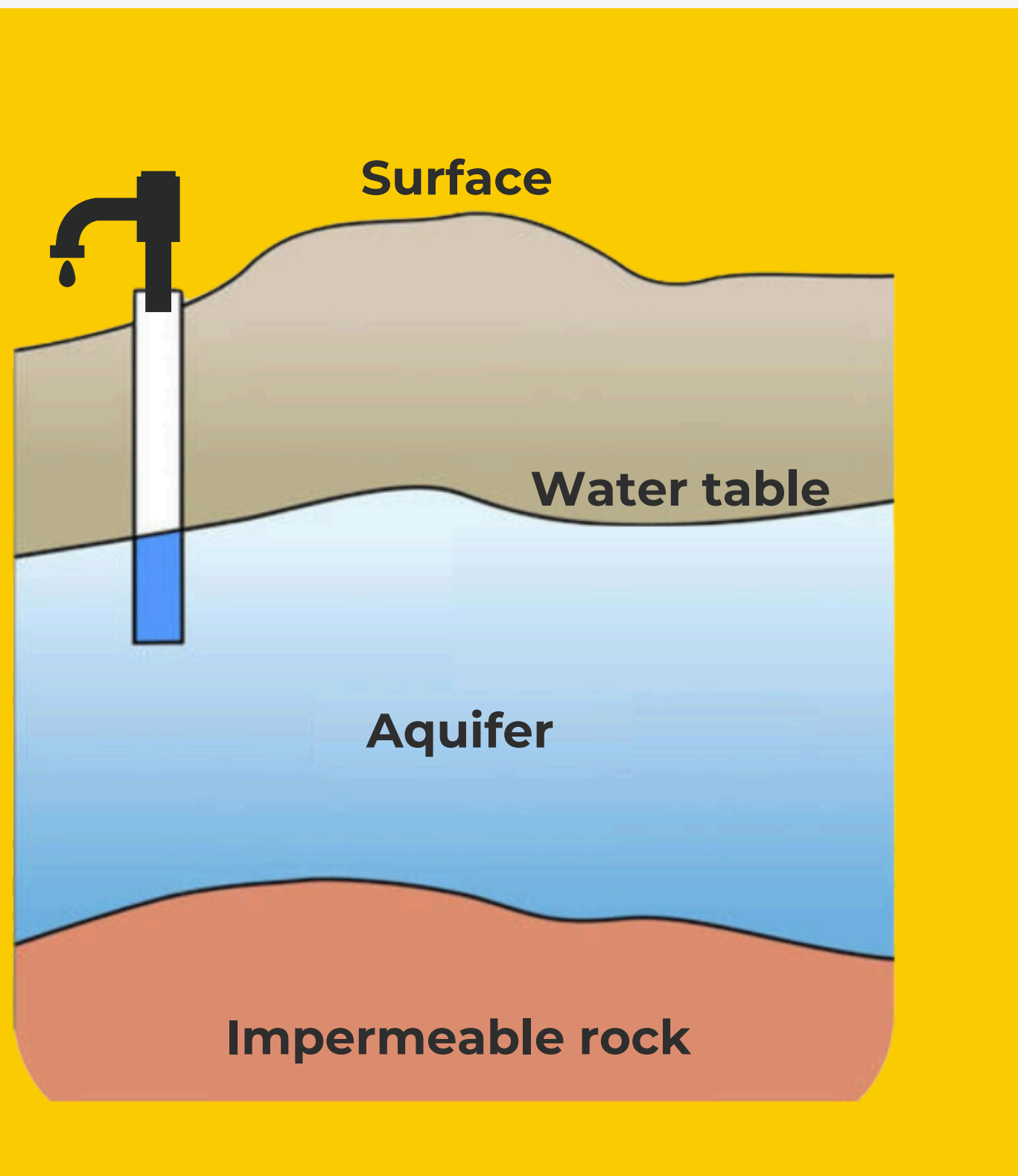
Rainwater and stormwater runoff can contribute to these water sources

3 Groundwater

Water that seeps into the ground, is collected from underground, and then stored in natural reservoirs called “aquifers”



Deep dive into Groundwater



“

Groundwater is the preferred water source for **rural water supplies across the Philippines**.

It is often **clear**, and somewhat free from contamination due to the **filtering effect of soil** on the water passing through it

”

- + Better quality than surface water
- + Less expensive to develop for use
- + Able to provide a more adequate supply in many areas in the country

Deep dive into Groundwater

How do we collect the groundwater if it is located underground?



Springs



Wells



Hand pumps

Things to Consider



Adequacy and reliability



Physical and chemical qualities



Costs



Legalities



Can you trust the quality of the water source? Will you be able to get water from it for a long time?

How much does it cost to source the water you use? Can your community pay for it for more than a year? Five years? Ten years?

Are you legally allowed to access and use that water source?



Topic 2

Water Distribution

What are the different levels?

Level I facility: Point Source



Used to describe settings where there is only a protected well or developed spring, but **no distribution network**

- Generally found in rural areas
- Normally serves an average of 15 households

Level II facility: Communal Faucet System / Stand Posts



Consists of a water source, storage, a piped distribution network, and communal faucets

- Generally found in rural and urban areas with clustered homes using a simple piped system
- Single faucet normally serves 4 to 6 households

Level III facility: Waterworks System / Individual House Connections



Has a source, a reservoir, a piped distribution network, and individual household taps

- Generally found in densely populated urban areas, where the population can afford individual connections

Why is water quality important and how to measure it?

Topic 3 Water Quality

Drinking Water

Quality Check



Clear



**Odorless, colorless, and
no unpleasant taste**



**Free from harmful
organisms, chemicals, and
radiological substances**

Water Quality

Parameters

“ Water quality parameters focus on three (3) aspects: ”



Physical Parameters



Chemical Parameters

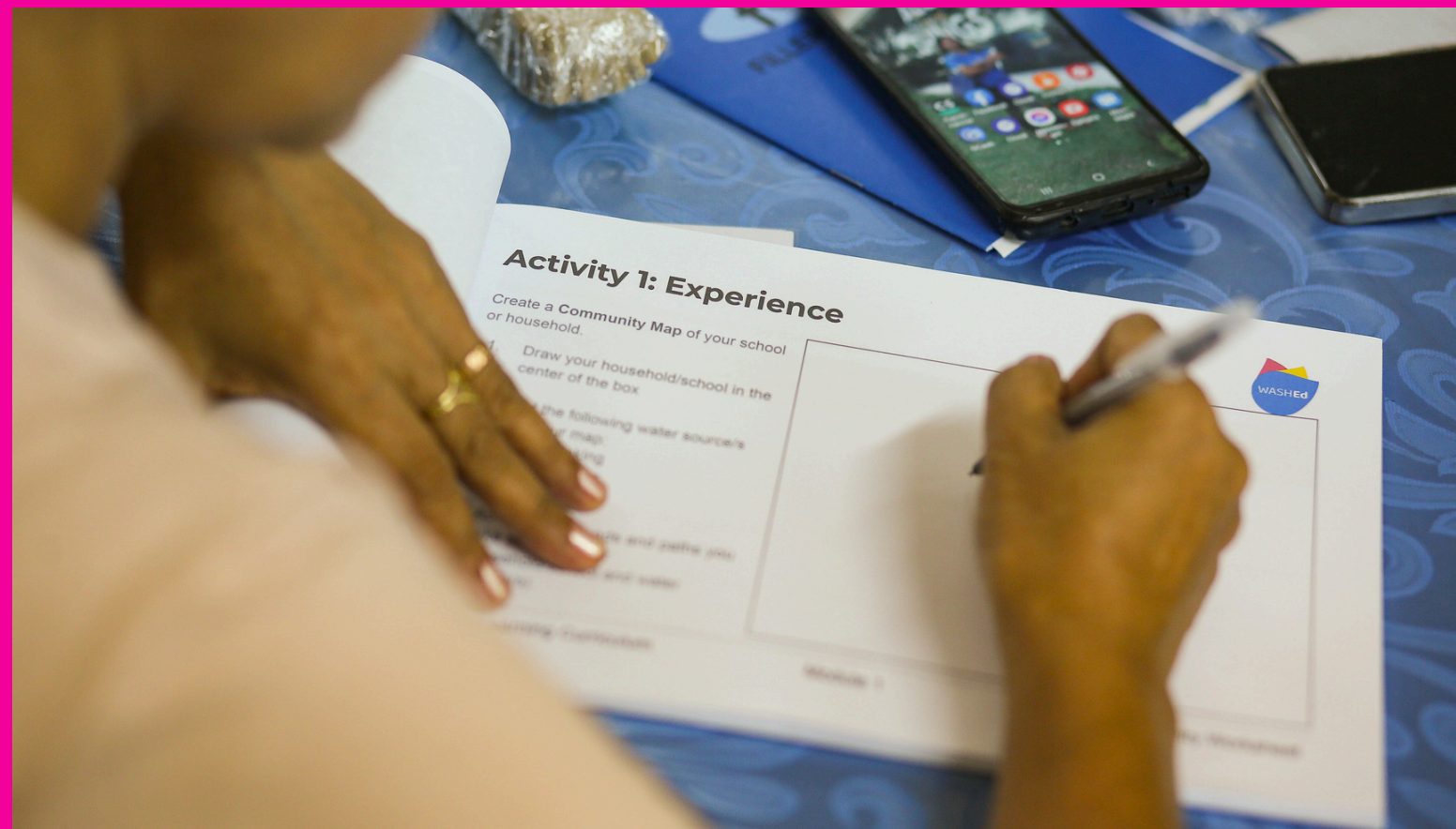


Biological Parameters

Activities

for Module 1

- Classroom activities are designed using the **E-SPOT Framework** to engage Active Learning
- **Think-Pair-Share** approach is recommended for all activities



Activity 1: Experience

Activity 2: Share

Activity 3: Process & Observe

Activity 4: Try



Digital files for the classroom activities will be available from the WASH-Ed Learning Portal



Activity 1: Experience

Create a **Community Map** of your school or household.

1. Draw your household/school in the centre of the box
2. Add the following water source/s into your map:
 - a. Drinking
 - b. Cooking
 - c. Washing
3. Where known, add the local roads and paths you take between your household/school and water source/s

A large, empty rectangular box with a black border, intended for drawing the Community Map.

Activity 2: Share



Assess each of your water source/s by completing the table below:

Water Usage	Classification (Level 1, Level 2, Level 3)	Reliability rating 1 = Being the lowest 5 = Being the highest	Reason for rating
DRINKING			
COOKING			
WASHING			
OTHER UTILITIES — 1 _____			
OTHER UTILITIES — 2 _____			

Activity 3: Process & Observe

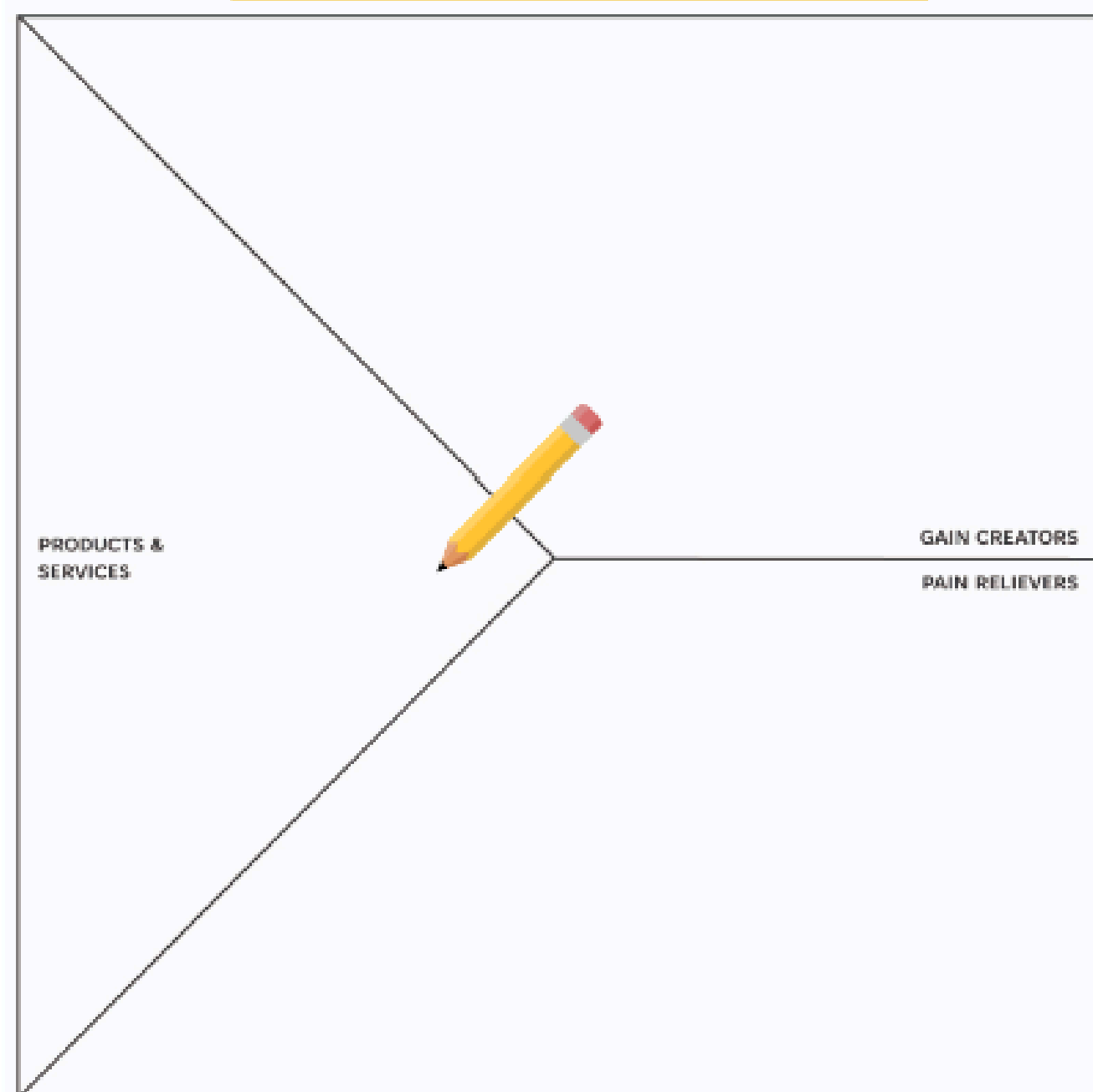


How does the reliability of these water sources impact your daily activities?

Activity 4: Try



VALUE MAP



USER PROFILE

